

Genetic report for longevity

Protein and Amino Acids

Amino acids in circulation (BCAA, glycine, phenylalanine, tyrosine, glutamine, histidine), PPM1K and the axis BCAA-mTOR, taurine (CSAD, SLC6A6, CDO1), creatine and methionine, blood proteins and protein intake

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The protein of the diet reaches the blood as amino acids. Some are signs: leucine and other branched-chain amino acids (BCAA) bind mTORC1, the nutrient sensor that rapamycin inhibits; glycine, serine and methionine feed methylation and glutathione; taurine, a cysteine derivative, low with age and its lack accelerated aging in animals (Science 2023). This report reads the circulating amino acid loci scores (from GWAS of nuclear magnetic resonance metabolites), PPM1K and other enzymes of BCAA catabolism, taurine, creatine and methionine genes, three polygenic blood protein scores (albumin, total protein, creatinine) and the relative protein intake score. With the retardation (less appetite), the objective of the ~140 g/day protein plan serves to protect lean mass; genes do not change this goal. The mTOR and autophagy are in the report Sensors of Nutrition and Autophagy, muscle in Muscle, Bone and Sarcopenia and methylation in the annex of the methylation pathway: here is only the bridge.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

12 variants	6 in the chip	6 imputed (DR2≥0.8)	0 not found	11 Scores with percentile
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Overall reading. A typical amino acid profile, with two points to note. BCAA and PPM1K: the 108 loci score of the BCAA is in the 35 (typical) percentile and the PPM1K rs1440581 T/C has a 1 copy of the allele that rises to the BCAA (typical); DBT C/C typical; **BCAT2 C/C** has 2 copies of the allele that rises to BCAA (unfavourable, small effect). There is no sign of slow BCAA catabolism that keeps the mTORC1 more connected. In the other amino acids (informative): glycine at the 47th percentile, glutamine in 41, tyrosine in 49, phenylalanine in 73 and histidine in 8. Taurine: the 2 loci score is at the 37th percentile (NLRP12 G/G); CSAD, CDO1 and SLC6A6 do not have common readable variants with known effect, so the genetics does not say if its taurine falls more or less with age. In blood proteins: albumin at the 36th percentile, total protein in 25 and creatinine in 9 (corresponds to the high renal filtration score of the Kidney and Uric Acid report). In ingestion: the relative protein score is in 82, but FGF21 rs838133 G/G does not have the most protein allele. The Health Report puts the metabolism of proteins as "WORSE RESPONSE" (the 82nd percentile), a label on the metabolic response to diets rich in protein; another old report of the same type appears as "Typical". None of this changes the objective of the ~140 g/day plan, which serves to protect lean mass with retatrutide.

Favourable

None.

Unfavourable

- **Protein metabolism (Health Report, 5 variant score):** percentile 82 (profile §23 (Health Report))
- **BCAT2 rs4801776 CC:** BCAT2: Mitochondrial transaminase of BCAA (first step of degradation).

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 6 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

PPM1K and BCAA catabolism (BCAA-mTOR axis)

PPM1K is the mitochondrial phosphatase that activates the BCKDH complex, the limiting step of the degradation of leucine, isoleucine and valine; BCAT2 makes the initial transamination and DBT is a subunit of BCKDH. Variants that lower PPM1K increase the circulating BCAA and, in Mendelian randomization, increase the risk of 2 type diabetes (Lotta 2016). High BCAA keeps the mTORC1 on; it is the route that rapamycin stops. Here the allele that rises the BCAA counts as unfavourable.

Ageing mechanisms: Deregulation of Nutritional Sensing; Insulin resistance; Autophagy / Deregulated Macroautophagy

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
BCAA: leucine, isoleucine and valine (GWAS loci)	Typical / intermediate	the 35th percentile · z=-0.38	Higher score = higher baseline BCAA (associated with insulin resistance and type 2 diabetes). Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 108 of 119 loci with genotype. Studies: Abar L (Isoleucine levels); Abar L (Leucine levels); Farashi S (Valine levels); Karjalainen MK (Isoleucine levels); Karjalainen MK (Leucine levels); Karjalainen MK (Valine levels); Richardson TG (Leucine levels); Richardson TG (Total concentration of branched-chain amino acids (leucine + isoleucine + valine)). Variants with genotype: 108 of 119 (covering of 79.5% coefficients).

Favourable

None.

Unfavourable

- BCAT2 rs4801776 CC

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
BCAT2	rs4801776 19:48800958 · C/T	CC	imputation DR2=0.95	C (2; 74%)	moderate	BCAT2: Mitochondrial transaminase of BCAA (first step of degradation). Meaning: GWAS Catalog: Valine levels (Zoodmsa M 2025; p=3e-105); T ↓ allele

Typical / intermediate variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
PPM1K	rs1440581 4:88305270 · T/A/C	TC	chip	C (1; 53%)	moderate	PPM1K rs1440581: the variant used in Mendelian randomization of BCAA (Lotta 2016); the allele that rises BCAA was associated with more diabetes type 2. Meaning: GWAS Catalog: Isoleucine levels (Lotta LA 2021; p=8e-37); C ↑ allele
DBT	rs2810424 1:100198123 · T/A/C	CC	imputation DR2=0.99	C (2; 92%)	moderate	DBT: E2 subunit of BCKDH complex. Meaning: GWAS Catalog: Leucine levels (Zoodmsa M 2025; p=5e-29); C ↑ allele

Other amino acids in circulation

Glycine (CPS1, GLDC: low glycine is associated with more coronary disease and diabetes), phenylalanine and tyrosine (PAH), glutamine (GLS2) and histidine. Informational loci scores: show the genetic trend for basal levels, not the effect of diet.

Ageing mechanisms: Metabolic Homeostasis decline; Redox imbalance

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Glycine (GWAS loci)	No judgement (informative)	the 47th percentile · z=-0.19	Informative: higher score = lower basal glycine. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 129 of 140 loci with genotype. Studies: Abar L (Glycine levels); Chang X (Glycine levels); Chen Y (Glycine levels); Farashi S (Glycine levels); Jia Q (Glycine levels); Karjalainen MK (Glycine levels); Richardson TG (Glycine levels); Wittemans LBL (Glycine levels). Variants with genotype: 129 of 140 (covering of 76.6% coefficients).
Phenylalanine (GWAS loci)	No judgement (informative)	the 73rd percentile · z=0.57	Information. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 31 of 33 loci with genotype. Studies: Farashi S (Phenylalanine levels); Karjalainen MK (Phenylalanine levels); Lotta LA (Phenylalanine levels); Richardson TG (Phenylalanine levels); Zoodma M (Phenylalanine levels). Variants with genotype: 31 of 33 (covering of 93.9% coefficients).
Tyrosine (GWAS loci)	No judgement (informative)	the 49th percentile · z=-0.04	Information. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 94 of 97 loci with genotype. Studies: Abar L (Tyrosine levels); Karjalainen MK (Tyrosine levels); Richardson TG (Tyrosine levels); Zoodma M (Tyrosine levels). Variants with genotype: 94 of 97 (covering of 96.6% coefficients).
Glutamine (GWAS loci)	No judgement (informative)	the 41st percentile · z=-0.23	Information. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 148 of 167 loci with genotype. Studies: Abar L (Glutamine levels); Farashi S (Glutamine levels); Karjalainen MK (Glutamine levels); Richardson TG (Glutamine levels); Zoodma M (Glutamine levels). Variants with genotype: 148 of 167 (covering of 88.2% coefficients).
Histidine (GWAS loci)	No judgement (informative)	the 8th percentile · z=-1.26	Information. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 51 of 57 loci with genotype. Studies: Abar L (Histidine levels); Farashi S (Histidine levels); Karjalainen MK (Histidine levels); Lotta LA (Histidine levels); Richardson TG (Histidine levels); Shin SY (Histidine levels); Zoodma M (Histidine levels). Variants with genotype: 51 of 57 (covering of 89.5% coefficients).

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CPS1	rs715 2:210678331 · T/A/C	TT	chip	C (0; 29%)	moderate	CPS1 T1406N: the largest locus of glycine (and homocysteine); the indicated allele rises glycine. Meaning: GWAS Catalog: Glycine levels (Wittemans LBL 2019; p=0e+00); C ↑ allele
GLDC	rs77979581 9:6542112 · T/A/C	TT	imputation DR2=0.97	T (2; 74%)	moderate	GLDC: glycine cleavage system; the indicated allele rises glycine. Meaning: GWAS Catalog: Glycine levels (Abar L 2024; p=9e-102); T ↑ allele
PAH		TC	imputation DR2=1.00	C (1; 58%)	moderate	PAH: converts phenylalanine into tyrosine; common variants move in basal

rs1814728
12:102854523
· T/C

phenylalanine. Meaning: GWAS Catalog:
Phenylalanine levels (Zoodmsa M 2025;
p=0e+00); C ↑ allele

GLS2	rs2657879 12:56471554 A/G/T	AG	chip	G (1; 21%)	moderate	GLS2: hepatic glutamines; the indicated allele rises to glutamine. Meaning: GWAS Catalog: Glutamine levels (Zoodmsa M 2025; p=0e+00); G ↑ allele
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Taurine: synthesis, transport and aging

Taurine is made from cysteine by cysteine dioxygenase (CDO1) and by decarboxylase of cisteinosulfinic acid (CSAD, which gives the hypotaurine), and enters the cells by TauT transporter (SLC6A6). In 2023, Singh et al. showed that taurine in the blood falls with age and that supplemental prolonged life and health in mice and monkeys (with observational data in humans). Plasma taurine GWAS has few loci (NLRP12, F12/GRK6); hypotaurine has a locus in CSAD, but the variant is rare and is not read with confidence. For SLC6A6 and CDO1 there are no common variants with replicated effect in GWAS; therefore, none of the three genes enter as SNP.

Ageing mechanisms: Mitochondrial dysfunction; Oxidative stress; Cell Senescence

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Plasma taurine (GWAS loci)	No judgement (informative)	the 37th percentile · z=-0.43	Informative; only 3 loci, very low power. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 2 of 2 loci with genotype. Studies: Calvo-Serra B (Taurine levels); Lotta LA (Taurine levels). Variants with genotype: 2 of 2 (covering of 100.0% coefficients).

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
NLRP12	rs4632248 · 19:53821741 · G/C/T	GG	imputation · DR2=0.98	T (0; 20%)	moderate	NLRP12: the strongest locus of plasma taurine; unknown mechanism. Meaning: GWAS Catalog: Taurine levels (Lotta LA 2021; p=1e-48); allele T ↑

Creatine, methionine and glycine-betaine

GATM (first step of creatine synthesis; the largest locus of creatine and creatinine), BHMT (remethyl to homocysteine with betaine, the TMG pathway of the plan) and MTHFR (5-MTHF of the plan; detail in the annex of the methylation pathway, without judgment here).

Ageing mechanisms: Epigenetic Changes; Thin Mass Loss

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MTHFR (C677T)	rs1801133 1:11796321 · G/A/C	AG	chip	—	strong	MTHFR C677T: detail and plan (5-MTHF 800 µg) in the annex of the methylation pathway.
BHMT (R239Q)	rs3733890 5:79126136 · G/A	AG	chip	—	weak	BHMT R239Q: common variant of betaine-homocysteine methyltransferase; small and inconsistent effect on homocysteine.
GATM	rs2486274 15:45374030 · G/C/T	GT	imputation DR2=1.00	T (1; 41%)	moderate	GATM: arginine-glycine amidintransferase, the first step of creatine (using glycine and arginine). Meaning: GWAS Catalog: Creatine levels (Surendran P 2022; p=2e-24); T ↑ allele

Blood proteins and protein intake

Polygenic scores of albumin, total protein and creatinine (creatinine reflects muscle mass and renal function) and the relative protein intake score (FGF21, the hepatic signal that regulates preference for protein and sugar). Information.

Ageing mechanisms: Thin Mass Loss; Sarcopenia; Metabolic Homeostasis Decline

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Albumin (60 423 variants)	No judgement (informative)	the 36th percentile · z=-0.30	Information. PGS001886 (PGS Catalog; Penalized regression (bigstatsr); Albumin; PGP000263, PMID 34995502). Variants with genotype: 58225 of 60423 (covering of 96.5% coefficients).
Total protein (69 557 variants)	No judgement (informative)	the 25th percentile · z=-0.67	Information. PGS002001 (PGS Catalog; Penalized regression (bigstatsr); Total protein; PGP000263, PMID 34995502). Variants with genotype: 66934 of 69557 (covering of 96.4% coefficients).
Creatinin (76 800 variants)	No judgement (informative)	the 9th percentile · z=-1.40	Informative: creatinine rises with muscle mass and decreases with renal filtration; the score mixes both things. PGS001945 (PGS Catalog; Penalized regression (bigstatsr); Creatinine; PGP000263, PMID 34995502). Variants with genotype: 73839 of 76800 (covering of 96.5% coefficients).
Relative protein intake (GWAS loci)	No judgement (informative)	the 82nd percentile · z=0.94	Informative: higher score = trend for more protein in the diet. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 5 of 7 loci with genotype. Studies: Merino J (Related protein intake). Variants with genotype: 5 of 7 (covering of 60.2% coefficients).
Protein metabolism (Health Report, 5 variant score)	Unfavourable	the 82nd percentile	It labels "WORSE RESPONSE" (metabolic response worse to diets rich in protein, in the reading of the original report). In another older report of the same type, the result appears as "Typical". Source: \$23 (Health Report) profile.

Favourable

None.

Unfavourable

- Protein metabolism (Health Report, 5 variant score) (the 82nd percentile)

Non-judgmental variants (informative) (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FGF21	rs838133 19:48756272 · A/C/G/T	GG	chip	—	moderate	FGF21 rs838133: allele A is associated with more protein and less carbohydrates in the diet (and more preference for sweet in other studies).

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Autophagy / Deregulated Macroautophagy	BCAT2 CC	—	BCAA: leucine, isoleucine and valine (GWAS loci) (the 35th percentile); PPM1K TC; DBT CC
Metabolic Homeostasis Decline	Protein metabolism (Health Report, 5 variant score) (the 82nd percentile)	—	—
Nutrient Sensing Deregulation	BCAT2 CC	—	BCAA: leucine, isoleucine and valine (GWAS loci) (the 35th percentile); PPM1K TC; DBT CC
Thin Mass Loss	Protein metabolism (Health Report, 5 variant score) (the 82nd percentile)	—	—
Insulin resistance	BCAT2 CC	—	BCAA: leucine, isoleucine and valine (GWAS loci) (the 35th percentile); PPM1K TC; DBT CC
Sarcopenia	Protein metabolism (Health Report, 5 variant score) (the 82nd percentile)	—	—
Epigenetic Changes	—	—	—
Redox imbalance	—	—	—
Mitochondrial dysfunction	—	—	—
Cell Senescence	—	—	—
Oxidative stress	—	—	—

Practical part

Ideas for protein and amino acids, crossed with the genetic plan v2 and current medication (retatrutide, rapamycin, dapagliflozin, telmisartan, tadalafil, lithium 5 mg, low dose naltrexone, isotretinoin). The current stack (creatine, TMG 3 g, glycine 5 g, 5-MTHF 800 µg, magnesium) remains as it is; everything is suggestion (not applied).

1. Keep ~140 g/day, distributed

Genetics: Albumin p36; total protein p25; creatinine p9; retatrutide

With the retatrutide cutting off the appetite, the protein is the first nutrient missing. Keep ~140 g/day in 3–4 meals of 30–40 g, the protein first in each meal, with a dose before sleeping. Sources: fish, eggs, fermented dairy products, soybeans and legumes.

V2 genetic plan and medication: Similar to the plan (protein ~140 g/day). • *suggestion (not applied)*

2. Food leucin, not isolated BCAA

Genetics: BCAT2 C/C; PPM1K T/C; BCAA–mTOR axis; rapamycin

The BCAA isolates bind the mTORC1 without the other amino acids and go in the opposite direction of rapamycin. The leucine of meals arrives for muscle synthesis, especially in strength training.

V2 genetic plan and medication: Plan compatible (BCAA isolates not recommended). • *suggestion (not applied)*

3. Taurine: discussing is not a priority

Genetics: Taurine p37 score (2 loci); CSAD/CDO1/SLC6A6 without readable variants

Taurine falls with age and, in animals, supplemental (equivalent to some grams per day) improved health and longevity; in humans there are only observational data and small trials. Fish and shellfish are the best sources. A supplement of 1–3 g/day is an option to discuss with the doctor; genetics does not give a special reason.

V2 genetic plan and medication: It is not in the plan; it would be an addition to discuss. • *suggestion (not applied)*

4. Creatine, glycine and TMG coherent with genes

Genetics: GATM G/T; glycine p47; BHMT and MTHFR without judgment

The creatine of the plan saves its own synthesis (GATM, which spends glycine and arginine), glycine 5 g and TMG 3 g feed the methylation and remethylation of homocysteine. Note for the analyses: creatine rises creatinine in blood without renal damage; cystatin C gives an estimate of filtration that is not affected.

V2 genetic plan and medication: The same as the plan. • *suggestion (not applied)*

Avoiding or with caution

- BCAA or leucine isolated. *suggestion (not applied)*
- To stay well below ~140 g/day on days of little appetite with the retatrutide. *suggestion (not applied)*
- To interpret a high creatinine as a renal problem without considering the creatine of the plan. *suggestion (not applied)*

Limitations

- The amino acid scores measure basal levels in circulation; they do not measure the protein needs, muscle synthesis or response to supplements.
- The GWAS of metabolites share loci with glucose, lipids and renal function (GCKR, CPS1, SLC2A4); part of each score reflects these traits.
- Taurine: GWAS has few loci, and the genes of synthesis and transport (CDO1, CSAD, SLC6A6) do not have common variants with replicated effect. The effect of taurine on aging in humans has not yet been tested in large trials.
- Hereditary errors in amino acid metabolism (phenylketonuria, urine disease in maple syrup, homocystinuria, GAMT/GATM deficits) are found in the Carrier State report when variants are readable.
- The percentiles compare with a reference population of 633 genomes.
- Only genes: does not include analysis (amino acids, albumin, creatinine, homocysteine). It is neither a diagnosis nor a prescription.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001886 (PGS Catalog; Penalized regression (bigstatsr); Albumin; PGP000263, PMID 34995502 · PGS002001 (PGS Catalog; Penalized regression (bigstatsr); Total protein; PGP000263, PMID 34995502 · PGS001945 (PGS Catalog; Penalized regression (bigstatsr); Creatinine; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

- Lotta LA et al. Genetic predisposition to an impaired metabolism of the branched-chain amino acids and risk of type 2 diabetes: a mendelian randomisation analysis (PLoS Med 2016; PMID 27898682).
- Singh P et al. Taurine deficiency as a driver of aging (Science 2023; PMID 37289866).
- Karjalainen MK et al. Genome-wide characterization of circulating metabolic biomarkers (Nature 2024).
- Wittemans LBL et al. Assessing the causal association of glycine with risk of cardio-metabolic diseases (Nat Common 2019).
- Merino J et al. Genome-wide meta-analysis of macronutrient intake of individual 91.114 (Mol Psychiatry 2019).
- Sørberg S et al. FGF21 is a sugar-induced hormone associated with sweet intake and preference in humans (Cell Metab 2017).
- PGS Catalog: PGS001886, PGS002001 and PGS001945 (Privé F et al., Am J Hum Genet 2022; PMID 34995502).
- GWAS Catalog (NHGRI-EBI), full download of associations.

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